Self-tuning Regulators under Disturbances (Åström & Wittenmark, Examples 3.9 and 3.10)

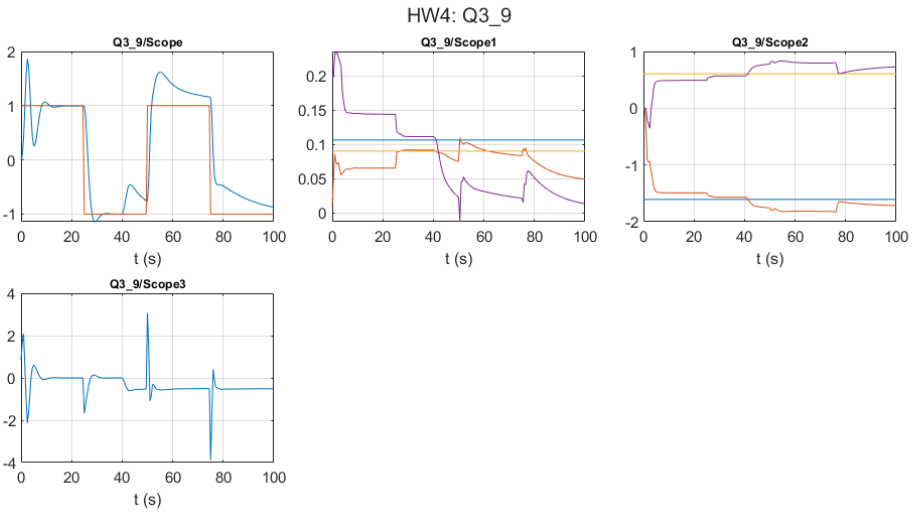
Adaptive Control, Homework 4

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Autumn 1404 (2025)

English summary of my original Persian report, which is kept as a PDF in this folder.

All cases are subsystems of `HW4.slx`. I compared three versions of the self-tuning regulator when a load disturbance acts on the plant.

1. **Standard STR (example 3.9).** Very sensitive to the disturbance. The noise enters the RLS estimator, so the estimates oscillate, the output has ripple, and the control signal is jumpy.
2. **STR with filtered signals u_f, y_f (example 3.10).** Filtering the regression signals gives the estimator cleaner data. The estimates oscillate less, the control is smoother and tracking improves, but the disturbance is not fully removed.
3. **STR with a disturbance model (example 3.10, modified).** The disturbance dynamics are included in the model, so the estimator learns them along with the plant. This gives the best tracking, the steadiest estimates and a smooth control signal, and is almost unaffected by the disturbance.



Re-run in R2023b: example 3.9, the standard STR under a disturbance.